

Checkpoint

12.2 Vectors

1 Problems

Problem 1. *What two things make up a vector?*

Problem 2. *What dimension does the following vectors live in and what is their sum?*
 $\vec{v} = \langle 2, 4, 9, -1 \rangle$ and $\vec{u} = \langle 0, -3, -3, 2 \rangle$

Problem 3. *Find the magnitude of $\langle 1, -1, 9 \rangle$ and find the magnitude of $-2\hat{i} + 5\hat{j} - 2\hat{k}$.*

Problem 4. *What is the unit vector going in the same direction as $2\hat{i} - \hat{j} + 3\hat{k}$?*

Problem 5. *What does it mean for two vectors to be parallel?*

2 Answers

1. a magnitude and a direction.
2. 4-dimensions $\vec{v} + \vec{u} = \langle 2, 1, 6, 1 \rangle$
3. $|\langle 1, -1, 9 \rangle| = \sqrt{83}$ and $|-2\hat{i} + 5\hat{j} - 2\hat{k}| = \sqrt{33}$
4. $\frac{2}{\sqrt{14}}\hat{i} - \frac{1}{\sqrt{14}}\hat{j} + \frac{3}{\sqrt{14}}\hat{k}$
5. They are scalar multiples. That is $\vec{u} = t\vec{v}$ where t is some real number.